Figure 2. Rafted holdfast epibionts on detached sub-Antarctic *Durvillaea* washed ashore in southern New Zealand.
Photo: J.M. Waters.

Where can I find out more?

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Migratory birds are lighter coloured

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Migratory birds undertake long and challenging journeys that have selected for a suite of adaptations from sensory mechanisms that facilitate orientation to extreme feats of endurance that push physiological limits. Recent work on two distantly related species revealed that migrating individuals increase their flight altitude dramatically during the day compared to at night[1,2]. These studies suggested that the phenomenon is driven by thermoregulation: the ascent to cooler heights during the day may offset heat generated by absorption of solar radiation. If thermoregulation is an important selective force on migratory species, migrants should have evolved lighter, more reflective plumage to avoid overheating. Here we show, across the entire avian radiation, that migratory species are indeed lighter coloured.

It has been hypothesised that animals benefit from being darker when living in cold regions and lighter where warm, because lighter colours absorb less solar radiation. This is known as the thermal melanism hypothesis, originally formulated for ectotherms, but also applying to endotherms, particularly birds[3]. If lighter-coloured plumage keeps birds cooler, then migratory species — especially long-distance migrants — should be lighter coloured than residents because differences in temperature between light and dark plumage persist despite the cooling effect of moving air[4]. To our knowledge only one study has reported that migratory birds are lighter coloured than residents, but it was based on a limited sample of 489 species of sexually monochromatic birds[5] and did not account for confounding factors (including climate).

To test the relationship between plumage lightness and migration, we quantified overall plumage lightness (from 0 = black to 100 = white, Supplemental information) for all species, averaging pixel lightness values from plates from the Handbook of the Birds of the World (https://birdsoftheworld.org), and aligned these data with the species' migratory behaviour[6]. Species were classified as either residents, short-distance migrants (migrate <2,000 km), variable migrants (some populations migrate more than >2,000 km and some less) and long-distance migrants (>2,000 km). We then tested the predicted effects of migration strategy on plumage lightness while controlling for factors known to affect plumage lightness — specifically climate, habitat structure and body size. Darker colours predominate in species from wetter regions and closed environments such as forests, most likely because they provide better camouflage in humid, shady and vegetated habitats[3]. In addition, darker plumage is more common in colder regions[3], and in smaller species[5]. Climatic data were obtained from Karger *et al.*[7] using distribution ranges from Birdlife International (http://datazone.birdlife.org), habitat openness was quantified based on habitat use data (http://datazone.birdlife.org), and body mass was obtained from Wilman *et al.*[8]. Because males and females can differ in colour[3], we analysed the data separately for each sex (species with complete information: $N_{females} = 9{,}754$, $N_{males} = 9{,}798$, Data S1A).

Phylogenetic comparative analyses revealed that, as predicted, resident species are on average darker than short-distance migrants ($\Delta_{females} = -1.16$, SE = 0.28, t = –4.14, p < 0.001, $\Delta_{males} = -0.75$, SE = 0.30, t = –2.45, p = 0.014), which were significantly darker than long-distance migrants ($\Delta_{females} = -1.22$, SE = 0.43, t = –2.84, p = 0.004; $\Delta_{males} = -1.37$, SE = 0.47, t = –2.91, p = 0.0036). Overall, average lightness increased from residents to long-distance migrants, both when controlling for the effects of potentially confounding covariates ($slope_{females} = 0.98$, SE = 0.11, t = 8.89, p < 0.001; $slope_{males} = 0.85$, SE = 0.12, t = 7.11, p < 0.001, Figure 1A), and without such control ($slope_{females} = 1.35$, SE = 0.113, t = 12.08, p < 0.001; $slope_{males} = 1.22$, SE = 0.12, t = 10.01, p < 0.001). Results were robust to different combinations of covariates (Data S1B). Migration effects are similar if we use migration distance (from Dufour *et al.*[6]) as a continuous variable rather than scores ($slope_{females} = 0.49$, SE = 0.08, t = 6.10; $slope_{males} = 0.44$,

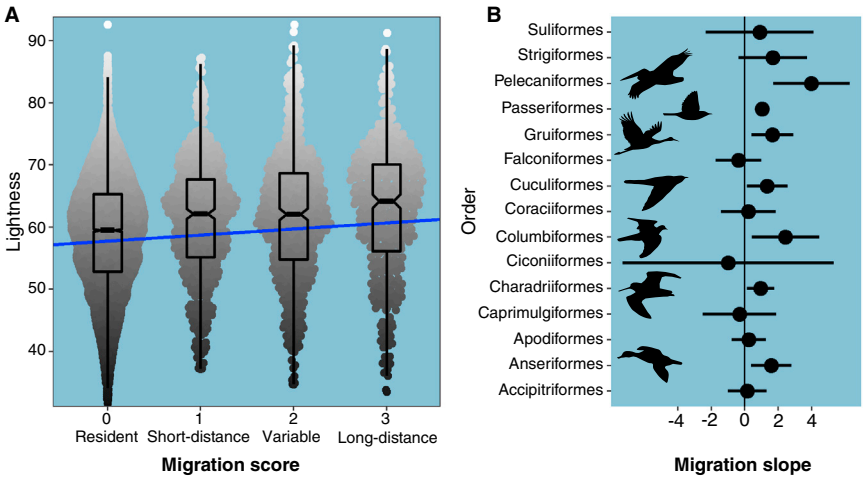

Figure 1. Migratory birds are lighter coloured than resident species.
(A) The distribution of average plumage lightness for females of each species in relation to the species' migration behaviour, classified as residents (score 0), short-distance migrants (score 1, <2,000 km migration distance), variable migration (score 2, some populations >2,000, some less) and long-distance migrants (score 3, >2,000 km). The blue line depicts the fitted line for all species (N = 9,754) after accounting for phylogenetic relatedness and confounding covariates (see Supplemental information), and boxplots represent median (middle line), 25th and 75th percentiles (box), and 1.5x interquartile range (whiskers) of the raw data. The notch constitutes an approximate 95% confidence interval for the median (1.58x interquartile range divided by the square-root of the sample size in each group). (B) Slopes and 95% confidence intervals depict the effects of migratory behaviour on female plumage lightness within 15 orders with both resident and long-distance migrants. Silhouettes illustrate bird orders with statistically significant effects.

SE = 0.087, t = 5.04; both $p < 0.001$). We also found that larger species were darker (contrary to Dunn *et al.*[5]) and that species that inhabited warm and dry environments with open vegetation were lighter (all effects $p < 0.001$, Table S1).

We further tested whether the observed effects were driven by one or a few highly speciose orders. We re-ran models (including confounding effects) separately for each of the 15 orders including both resident species and long-distance migrants. These revealed the predicted positive slope (lighter plumage in migrants) in 12 orders and statistically significant effects in 7, including taxonomically disparate birds such as passerines, cuckoos, ducks and geese, shorebirds, pigeons, pelicans, and cranes (Figure 1B). Three orders had negative slopes that were not statistically significant (Figure 1B).

Although the effects of migration had strong statistical support, many species did not follow the general pattern, as illustrated by the wide range of lightness values in each migration category (Figure 1A). Multiple factors affect the evolution of plumage colour[5], and light-coloured plumage showed a strong phylogenetic signal ($\lambda > 0.88$), indicating evolutionary constraints. It has been suggested that migratory birds should have darker upperparts, because warmer dorsal surfaces reduce friction and thereby increase flight efficiency[4,9]. Analyses on a large order of species for which we had data on dorsal and ventral lightness (Passeriformes)[3] confirm that migrants are lighter dorsally and ventrally. However, the effect is weaker for the generally darker dorsal plumage (Data S1C), which could indicate a trade-off between thermoregulation and flight efficiency. Colour is one of many potential adaptive responses to the thermoregulatory needs of migratory species, in addition to variation in body size (Bergmann's rule[10]) and bill size (Allen's rule[10]), or the described adjustments in migratory altitude[1,2]. Future studies should determine how these interact to shape the optimal migratory phenotype. For example, we hypothesize that darker migratory species show more flexibility in migratory altitude (that is, they ascend higher during the day) than lighter ones, or that they will favour nocturnal rather than diurnal migration.

In conclusion, we demonstrate a general association between lightness and migratory behaviour across all birds. Our results confirm previous findings reporting a correlation between plumage lightness and migration[5]. The authors speculated that lighter plumage in migratory birds may be used in competition for territories. Instead, we suggest thermoregulation as an alternative explanation that fits with the reported changes in migration altitude between night and day[1,2].

SUPPLEMENTAL INFORMATION

Supplemental information includes one table, results, experimental procedures, author contributions, references, and one data file, and can be found with this article online at https://doi.org/10.1016/j.cub.2021.10.048.

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